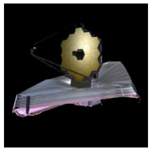

8507 - Confirming sub-Jupiter mass planets at 20 au around a nearby star

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Tiffany Meshkat (PI)	**California Institute of Technology**
Dr. Charles A. Beichman (CoI)	Jet Propulsion Laboratory
William Balmer (CoI)	Space Telescope Science Institute
Dr. Geoffrey Bryden (CoI)	Jet Propulsion Laboratory
Julien Girard (CoI)	Space Telescope Science Institute
Dr. Jarron Leisenring (CoI)	University of Arizona
Jorge Llop-Sayson (CoI)	Jet Propulsion Laboratory
Dr. Marshall Perrin (CoI)	Space Telescope Science Institute
Dr. Laurent Pueyo (CoI)	Space Telescope Science Institute
Dr. Marie Ygouf (CoI)	Jet Propulsion Laboratory
Dr. Alexandra Greenbaum (CoI)	California Institute of Technology
Rohan Kane (CoI)	Space Telescope Science Institute

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Altair				
	1	Altair - Roll 1	NIRCam Coronagraphic Imaging	(1) ALTAIR
	2	Altair - Roll 2	NIRCam Coronagraphic Imaging	(1) ALTAIR
	3	Vega - Altair Reference star	NIRCam Coronagraphic Imaging	(2) VEGA

ABSTRACT

Altair is one of our nearest neighbors (5.13 pc) and its age has recently been revised to a much lower 88 Myr. It is now among the youngest nearby stars, making it an ideal candidate for exoplanet direct imaging searches with JWST. Cycle-3 observations of Altair have revealed several exciting candidate companions. The closest candidate companion to Altair is at a projected separation of about 4”, with a contrast of 17.4 mag. Assuming an age of 88 Myr, this candidate corresponds to a mass of 0.5 Jupiter masses at a separation of 20 au. We propose to obtain follow-up images of Altair with NIRCam to confirm the planetary nature of the candidate companions to the North of the star. Due to the very high proper motion of Altair, we will be able to confirm if any of the candidate companions are co-moving with Altair even at the start of the Cycle-4 window. If confirmed, this would be the lowest mass planet ever imaged around a star and would be the target of significant follow-up for characterization.

OBSERVING DESCRIPTION

The proposed follow-up images of Altair will make simultaneous long and short wavelength filter observations using the M430R mask in subarray mode. Following the observations of cycle-3, this program will include F444W and simultaneously F200W. The F200W filter will assist in rejection of background objects. Unlike cycle-3, we will schedule the observations to occur approximately 6 months or 18 months after the previous data were obtained, in order to observe the North side of Altair. We will use the same integration times as Cycle-3.

The star will be observed at two roll angles to enable Angular Differential Imaging (ADI) and a reference star (Vega) will be observed for Reference Differential Imaging (RDI), with 5 dithers to introduce PSF diversity.

Due to the brightness of these stars, target acquisition requires RAPID integrations, even after dimming with a neutral density filter. The shortest integration time yields an SNR greater than 300.

All science observations use the F430R NIRCam coronagraphic mask, with the F200W and F444W filters. For these bright stars, the RAPID readout mode is employed, allowing up to 10 groups of observations before saturation occurs. The total exposure time for the reference star, when the 5 dithers are combined, matches the exposure time for each roll angle of the primary target. We have selected the dates to observe at a time when the Northern side of Altair is visible.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ALTAIR	RA: 19 50 46.9985 (297.6958271d) Dec: +08 52 5.96 (8.86832d) Equinox: J2000	Proper Motion RA: 536.23 mas/yr Proper Motion Dec: 385.29 mas/yr Parallax: 0.19495" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[A stars] Extended=NO				
(2)	VEGA	RA: 18 36 56.3364 (279.2347350d) Dec: +38 47 1.28 (38.78369d) Equinox: J2000	Proper Motion RA: 200.94 mas/yr Proper Motion Dec: 286.23 mas/yr Parallax: 0.130" Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[A stars, Debris disks] Extended=NO				

Proposal 8507 - Observation 1 - Confirming sub-Jupiter mass planets at 20 au around a nearby star

Observation

Proposal 8507, Observation 1: Altair - Roll 1

Thu Apr 03 18:00:14 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Altair - Roll 1 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ALTAIR	RA: 19 50 46.9985 (297.6958271d) Dec: +08 52 5.96 (8.86832d) Equinox: J2000	Proper Motion RA: 536.23 mas/yr Proper Motion Dec: 385.29 mas/yr Parallax: 0.19495" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[A stars]
Extended=NO

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F335M	BRIGHT (ND Square)	RAPID	9	1	1	0.504	157600.4

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK430R	true	SUB320A430R	NONE

Confirmation

#	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time	Conf. Total Dithers
1	RAPID	5	2	2	118.104	1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F200W	F444W	RAPID	5	650	1	650	4182.568	157600.6

PSF References	Vega - Altair Reference star (Obs 3) (PSF Reference; Filters [F200W/F444W]) Additional Justification: false
Special Requirements	Between Dates 01-MAY-2025:00:00:00 and 31-MAY-2025:00:00:00 Offset 0.0 arcsec, -0.016 arcsec No Parallel Attachments Group Observations 1, 2, 3, Non-interruptible Aperture PA Offset 1 from 2 by 8 to 14 Degrees (Same offsets in V3)

Observation

Proposal 8507, Observation 2: Altair - Roll 2

Thu Apr 03 18:00:14 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Altair - Roll 2 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.

(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	ALTAIR	RA: 19 50 46.9985 (297.6958271d) Dec: +08 52 5.96 (8.86832d) Equinox: J2000	Proper Motion RA: 536.23 mas/yr Proper Motion Dec: 385.29 mas/yr Parallax: 0.19495" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[A stars]
Extended=NO

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F335M	BRIGHT (ND Square)	RAPID	9	1	1	0.504	157600.4

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK430R	false	SUB320A430R	NONE

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F200W	F444W	RAPID	5	650	1	650	4182.568	157600.6

PSF References

Vega - Altair Reference star (Obs 3) (PSF Reference; Filters [F200W/F444W])

Additional Justification: false

Special Requirements	
	Offset 0.0 arcsec, -0.016 arcsec Group Observations 1, 2, 3, Non-interruptible Aperture PA Offset 1 from 2 by 8 to 14 Degrees (Same offsets in V3)

Observation

Proposal 8507, Observation 3: Vega - Altair Reference star

Thu Apr 03 18:00:14 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	VEGA	RA: 18 36 56.3364 (279.2347350d) Dec: +38 47 1.28 (38.78369d) Equinox: J2000	Proper Motion RA: 200.94 mas/yr Proper Motion Dec: 286.23 mas/yr Parallax: 0.130" Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[A stars, Debris disks]
Extended=NO

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	F335M	BRIGHT (ND Square)	RAPID	9	1	1	0.504	157600.5

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK430R	false	SUB320A430R	5-POINT-BOX

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	F200W	F444W	RAPID	5	140	5	700	4504.304	157600.7

PSF References

PSF Reference: true

Special Requirements	
	Offset 0.0 arcsec, -0.016 arcsec Group Observations 1, 2, 3, Non-interruptible